



Customer Behaviour Analysis

From raw retail data to actionable business insights — a complete end-to-end analysis using Python, SQL, and Power BI.

About the Dataset



Type

Retail / E-commerce — Structured Tabular Data

Level

Customer Transaction Level

Numerical

Age, Purchase Amount, Previous Purchases

Categorical

Category, Payment Method, Season

Used for **customer behavior analysis** and **business decision-making**.

Why This Data Matters



High-Value Customers

Identify and retain top spenders



Purchase Patterns

Understand buying frequency and habits



Discount Strategy

Optimize promo effectiveness



Customer Lifetime Value

Increase long-term revenue per customer



The Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve retention, and optimize marketing?

Low Retention

Customers not purchasing frequently — who are repeat vs. one-time buyers?

Ineffective Campaigns

Do discounts drive purchases? Which customers respond to promo codes?

Product Demand

Which categories perform best and generate the highest revenue?

Customer Segmentation

Identify high-value, frequent, and discount-sensitive customers

Data Dictionary

Customer ID	Unique customer identifier
Age / Gender	Customer demographics
Item / Category	Product and category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent (USD)
Season / Location	Purchase season and geographic location
Review Rating	Customer product rating
Subscription Status	Subscribed (Yes/No)
Discount Applied	Whether a discount was used
Previous Purchases	Number of past purchases
Payment Method	Cash, Credit Card, Debit Card, etc.

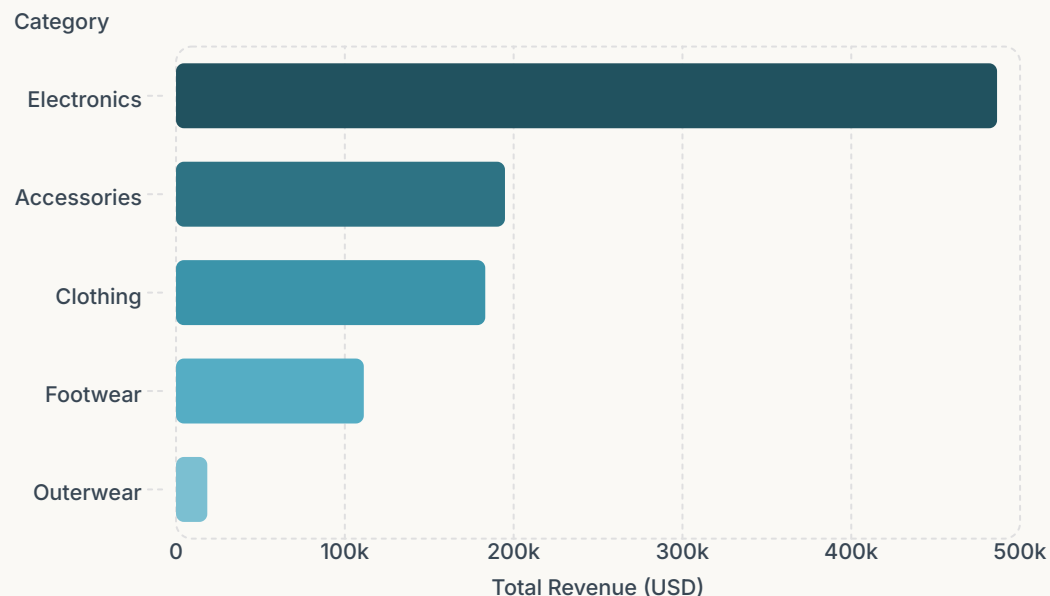
EDA — Data Cleaning in Python



Key techniques: category-specific imputation (e.g., Electronics → "Not Applicable"), product-level mean for ratings, zero-fill for Previous Purchases, and duplicate removal by Customer ID. Column names standardized to lowercase with underscores for SQL/Power BI compatibility.

SQL Analysis — Revenue Insights

Revenue by Category



Key Findings

Discounts & Spending

Discounted orders have higher avg. spend (\$219 vs \$182), but lower total revenue

Subscribers Spend More

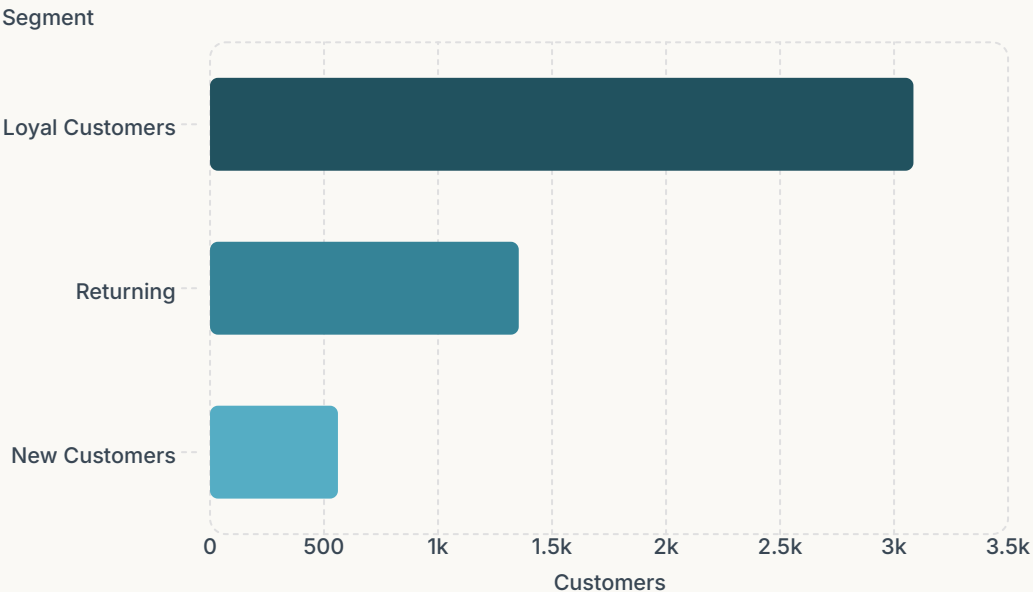
Avg. spend: \$270 (subscribers) vs \$165 (non-subscribers)

Gender Revenue

Male: \$390K · Female: \$352K · Other: \$251K

SQL Analysis — Customer & Product Insights

Customer Segments



Top Products by Review Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79



Lowest-rated: Phone (2.94), Headphones (3.08), Watch (3.10)

Power BI Dashboard — Key Metrics

5000

Total Customers

30

Unique Items

3.62

Avg Rating

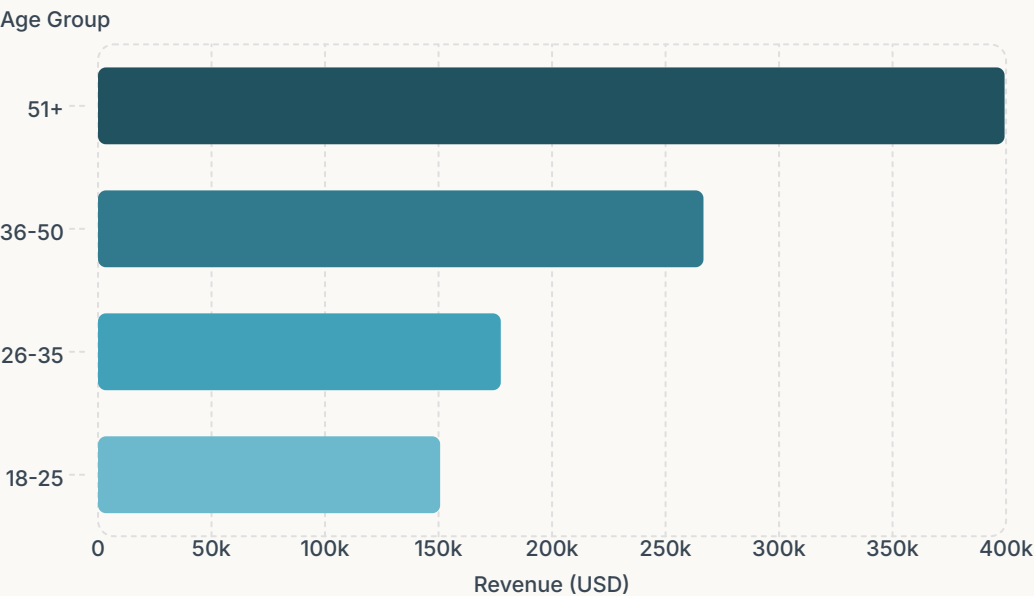
994K

Total Spend (USD)

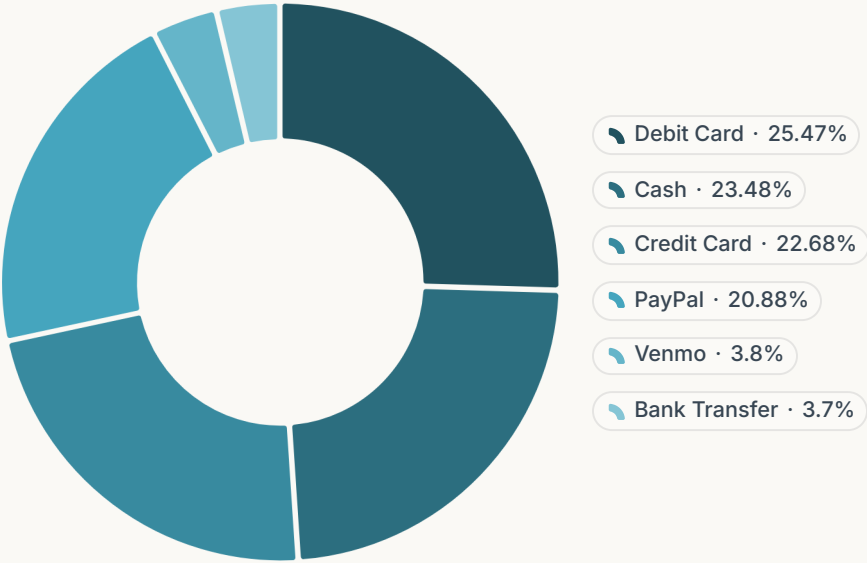
198

Avg Spend (USD)

Revenue by Age Group



Payment Methods





Key Takeaways

1 Electronics Dominates Revenue

\$486K — nearly 2.5× the next category. Focus inventory and marketing here.

2 Subscribers Are High-Value

63% higher average spend. Prioritize subscription conversion campaigns.

3 51+ Age Group Leads Spending

40% of total revenue. Tailor product recommendations to this segment.

4 Loyal Customers = 62% of Base

3,085 loyal customers. Invest in retention programs to protect this segment.